

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Course Code: ITA0609

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CO2 ASSESSMENT TOOL 3 - SCENARIO BASED QUESTIONS

1. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

Introduction

Backpropagation is a supervised learning algorithm used to train a Multilayer Perceptron (MLP). It minimizes prediction error by updating the weights using gradient descent. The algorithm consists of three stages: forward pass, error calculation, and backward propagation.

Simple Example

Consider a neural network with:

- **Inputs:** $x_1 = 1, x_2 = 0$
- **Target Output (T):** 1
- **Initial Weights:** $w_1 = 0.5, w_2 = 0.4$
- **Bias:** $b = 0.1$
- **Learning Rate (η):** 0.1
- **Activation Function:** Sigmoid

Step 1: Forward Pass

Weighted Sum

$$\begin{aligned} z &= x_1 w_1 + x_2 w_2 + b \\ &= 1(0.5) + 0(0.4) + 0.1 = 0.6 \end{aligned}$$

Sigmoid Activation

$$y = \frac{1}{1 + e^{-z}}$$
$$= \frac{1}{1 + e^{-0.6}} = 0.646$$

Predicted Output = 0.646

Step 2: Error Calculation

Target Output = 1

Prediction = 0.646

Error

$$Error = T - y$$
$$= 1 - 0.646 = 0.354$$

Mean Squared Error

$$E = \frac{1}{2}(T - y)^2$$
$$= \frac{1}{2}(0.354)^2 = 0.0625$$

Step 3: Backward Propagation

Gradient:

$$\delta = (T - y) \times y(1 - y)$$
$$= 0.354 \times 0.646 \times 0.354 = 0.081$$

Update Weight

$$w_{new} = w + \eta \delta x$$

For w_1

$$w_1 = 0.5 + 0.1(0.081)(1) = 0.5081$$

For w_2

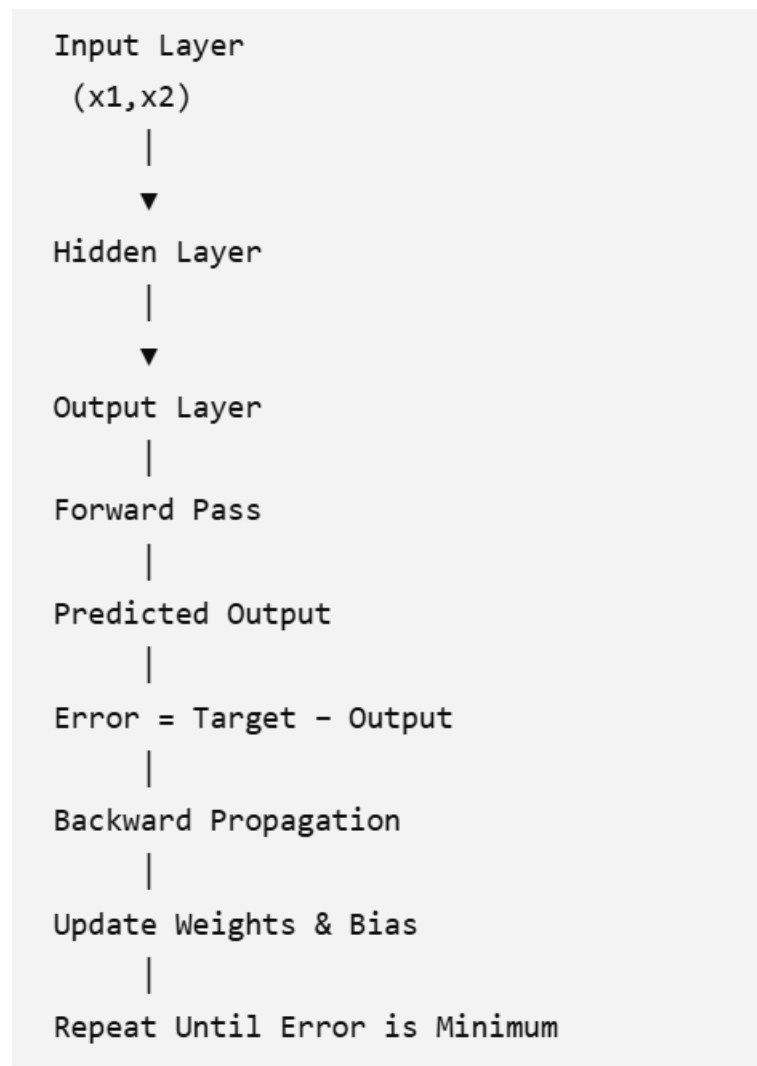
$$w_2 = 0.4 + 0.1(0.081)(0) = 0.4$$

Updated Bias

$$b = 0.1 + 0.1(0.081) = 0.1081$$

The updated weights reduce the prediction error in the next iteration.

Backpropagation Process



Effect of Learning Rate

- Determines the size of weight updates.
- Small learning rate → Slow but stable convergence.
- Large learning rate → Faster learning but may overshoot the optimum.
- An optimal learning rate provides faster and stable convergence.

2. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution.

Dataset

Candidate	Speed	Accuracy	Suitable
A	70	80	Yes
B	60	75	No
C	90	85	Yes
D	65	70	No

Fitness Function

$$\text{Fitness} = \text{Speed} + \text{Accuracy}$$

Generation 0 (Initial Population)

Candidate	Fitness
A	150
B	135
C	175
D	135

Best Candidate: C (Fitness = 175)

Step 1: Selection

The two individuals with the highest fitness are selected.

Parents:

- C (90, 85)
- A (70, 80)

These are chosen because they have the highest fitness values.

Step 2: Crossover

Single-point crossover combines parent features.

Parent 1

Speed = 90

Accuracy = 85

Parent 2

Speed = 70

Accuracy = 80

Offspring

Offspring	Speed	Accuracy
O ₁	90	80
O ₂	70	85

Step 3: Mutation

Randomly modify one gene.

Suppose O₁ speed changes:

90 → 92

Mutated Offspring:

Offspring	Speed	Accuracy
O ₁	92	80

Fitness

$$92 + 80 = 172$$

Generation 1

Candidate	Speed	Accuracy	Fitness
C	90	85	175
O ₁	92	80	172
O ₂	70	85	155

Generation 2

Apply another crossover and small mutation.

New Candidate

Speed = 92

Accuracy = 88

Fitness

$$92 + 88 = 180$$

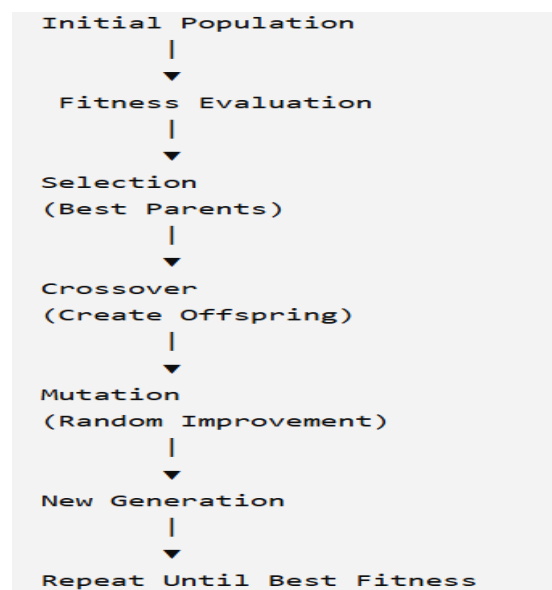
This fitness is greater than all previous candidates.

Final Optimized Solution

Candidate	Speed	Accuracy	Fitness
Optimized Solution	92	88	180

Suitable = Yes

Evolution Process



Role of Genetic Algorithm Operations

1. Selection

- Chooses individuals with the highest fitness.
- Ensures strong candidates pass their genes to the next generation.
- Improves solution quality over generations.

2. Crossover

- Combines characteristics of two parents.
- Produces new offspring with potentially better performance.
- Increases diversity while preserving good traits.

3. Mutation

- Randomly changes one or more genes.
- Prevents the algorithm from getting stuck in local optima.
- Explores new regions of the search space and maintains genetic diversity.

Conclusion

The Genetic Algorithm starts with an initial population, selects the fittest candidates, generates new offspring through crossover, and introduces random mutations to explore better solutions. After successive generations, the algorithm converges to the **optimal candidate (Speed = 92, Accuracy = 88, Fitness = 180)**, demonstrating how evolutionary operations improve solution quality over time.

3.

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

ANSWER

Step 1: Identify Actual and Predicted Classes

Assume "Yes" = Positive and "No" = Negative.

Example	Actual Output	Predicted Output	Result
1	Yes	Yes	True Positive (TP)
2	Yes	No	False Negative (FN)
3	No	No	True Negative (TN)
4	Yes	Yes	True Positive (TP)
5	No	Yes	False Positive (FP)

Step 2: Confusion Matrix Values

- True Positives (TP) = 2
- True Negatives (TN) = 1
- False Positives (FP) = 1
- False Negatives (FN) = 1

Confusion Matrix

Predicted Yes Predicted No

Actual Yes TP = 2 FN = 1

Actual No FP = 1 TN = 1

Step 3: Performance Evaluation Metrics

1. Accuracy

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ &= \frac{2 + 1}{2 + 1 + 1 + 1} = \frac{3}{5} = 0.6\end{aligned}$$

Accuracy = 60%

2. Precision

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{2}{2 + 1} = \frac{2}{3} = 0.667\end{aligned}$$

Precision = 66.7%

3. Recall (Sensitivity)

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP + FN} \\ &= \frac{2}{2 + 1} = \frac{2}{3} = 0.667\end{aligned}$$

Recall = 66.7%

4. F1-Score

$$\begin{aligned}F1 &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= \frac{2 \times 0.667 \times 0.667}{0.667 + 0.667} = 0.667\end{aligned}$$

F1-Score = 66.7%

5. Specificity

$$\begin{aligned}\text{Specificity} &= \frac{TN}{TN + FP} \\ &= \frac{1}{1 + 1} = \frac{1}{2} = 0.5\end{aligned}$$

Specificity = 50%

Step 4: Performance Analysis

- **Accuracy (60%)** indicates that the model correctly classified **3 out of 5** samples.
- **Precision (66.7%)** means that when the model predicts "Yes," it is correct about **67%** of the time.
- **Recall (66.7%)** shows that the model correctly identifies about **67%** of the actual positive cases.
- **F1-Score (66.7%)** indicates a moderate balance between precision and recall.
- **Specificity (50%)** shows that the model correctly identifies only half of the actual negative cases.

Overall-Analysis:

The neural network demonstrates **moderate performance**. It successfully detects most positive cases but still produces one false positive and one false negative. Improving preprocessing, increasing training data, and selecting more informative features could improve its predictive performance.

Step 5: Role of Preprocessing

Preprocessing improves the quality of input data before training.

- Removes missing or inconsistent values.
- Encodes categorical variables (High, Medium, Low) into numerical form.
- Normalizes numerical features like Age and Glucose Level.
- Removes duplicate or noisy records.
- Balances class distribution if the dataset is imbalanced.

Benefits

- Faster model convergence.
- Reduced overfitting.
- Better prediction accuracy.
- Improved reliability and consistency.

Step 6: Role of Feature Selection

Feature selection identifies the most relevant input variables.

- Removes irrelevant or redundant features.
- Reduces computational complexity.
- Improves model interpretability.
- Decreases the risk of overfitting.
- Enhances generalization to unseen data.

Benefits

- Higher accuracy.
- Faster training.
- Better robustness.
- More reliable predictions.